

THE MINISTRY OF EDUCATION OF THE RUSSIAN FEDERATION
FEDERAL STATE BUDGETARY EDUCATIONAL INSTITUTION OF
HIGHER EDUCATION
«RUSSIAN STATE PEDAGOGICAL UNIVERSITY IN THE NAME OF A. I.
HERZEN»



Computer Science and Engineering 09.03.01 Software Development Technology

Domain of study «Software Development Technology»

Final qualifying work

«Modeling Fractals»

Student of the 4-th course
in full time study
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Saint Petersburg
2024

INTRODUCTION

Relevance of the topic

What do a tree, a seashore, a cloud or the blood vessels in our hand have in common? At first glance, it may seem that all these objects have nothing in common. However, in fact, there is one structural property inherent in all of the listed objects: they are self-similar. From a branch, as from a tree trunk, smaller shoots extend, from them - even smaller ones, and so on, that is, the branch is similar to the whole tree. In this, they are similar to one of the most beautiful mathematical objects - fractals.

Fractals are among the most amazing and multifaceted objects in mathematics. Their unique properties, such as self-similarity and fractional dimensionality, find wide application in various fields of science and technology. Fractals can be found in nature, for example, in the shapes of mountains, clouds, plants and coastlines. In addition, they are widely used in computer graphics to create complex and realistic images, as well as in the modeling of dynamic systems and the analysis of financial markets. The relevance of studying fractals is due to their ability to model complex systems and processes, which makes them an important tool for modern research and technology.

Purpose and Objectives

The purpose of this final qualifying work is to study the methods of fractal modeling. To achieve this goal, the following tasks were set:

1. Explore the properties of fractals and their mathematical foundations.
2. Consider the types of fractals.
3. Analyze different methods of fractal modeling.
4. Implement fractal models using Python programming language.

Object of research: fractal structures and their properties.